



SDEV2501

Day 01 - Intro to Data Science & Analytics

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Today's objective

- Define and understand key concepts in data science & data visualization

Agenda

- Introductions
- Housekeeping & Classroom Expectations
- NAIT Resources
- Course Orientation
- Intro to Data Science

INTRODUCTIONS

**WE
ARE** ESSENTIAL
TO ALBERTA



About your instructor

About you

Share as much as is comfortable!

e.g.

- ❑ How you're comfy being addressed
- ❑ Your hometown
- ❑ Hobbies, interests, favourite sports or movies, etc
- ❑ If you have kids or pets

HOUSEKEEPING & CLASSROOM EXPECTATIONS

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Housekeeping: Attendance

- You are expected to attend class.
 - Please let your instructor know if you are unable to attend a class. You are responsible for catching up on any missed material.
- Punctuality is appreciated.
 - But I'd rather you join late than not at all.
- Breaks will happen every ~45 min.
 - Let me know if you need a break sooner!
- Interruptions are disruptive.
 - Make sure your devices are set to silent.
 - If you need to step out for any reason, please try to do so quietly.

Housekeeping: Communication

- **Questions** are very welcomed!
 - During class: raise your hand (or use chat for online classes).
 - Outside of class: use Teams channel or email.
 - If questions are out of scope (or we're short on time) I may add them to the Parking Lot.
- I strive to respond to **emails & Teams messages** within 1-2 business days.
 - For emails, a clear subject line usually means a quicker response.
 - e.g. "SDEV 2501 Assignment 1" vs "homework"

Housekeeping: In-Person Logistics

- Nearest exit
- WCs
 - Men's: A and B wings
 - Women's: B and C wings
 - Gender-inclusive: 1st floor (W106)

ASSESSMENTS

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Assessment Expectations

- Students are responsible for knowing and adhering to due dates for assessments.
- In this course, generative AI is not permitted for assessments, as they are intended to measure your ability to generate your own work. Using prohibited resources will be considered academic misconduct and be addressed following SR 1.3 Academic Integrity Procedure.
- Instructors strive to provide marks & feedback within 5 business days.

Assignment **Extension** Guideline

- Students with **accommodations** are responsible for disclosing those requirements at least 10 days in advance of any extension request.
- In the case of extensions requested due to **extenuating circumstances** (e.g. medical emergency), notify your instructor as soon as possible, 10 days in advance where possible.
 - Documentation is generally required to support the request.
 - The request should be made in writing and sent via email.
- For requests due to **religious observances**, submit a [Deferred Exam Request Form](#) at least 10 business days prior.

NAIT RESOURCES

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NAIT Resources

- [Tutoring & accommodations](#)
- [Student rights & responsibilities](#)
- [Counselling](#)
- [Mental health](#)
- [Spiritual well-being](#)
- [International centre](#)
- [Nîsôhkamâtôtân Centre](#)
- [2SLGBTQIA+ Students](#)
- [Scholarships](#)
- [Financial Aid](#)
- [Job postings](#)
- [NAITSA events](#)

COURSE ORIENTATION

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Course Orientation

- Course outline
- Syllabus
- Course calendar
- BrightSpace tour
- What will we be able to do by the end of this course?

DATA ANALYSIS

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What is data analysis?

Data analytics: using analyzed data to make data-driven decisions

- Data **wrangling**: cleaning and transforming raw data into a better format
- Data **analysis**: collecting, manipulating, examining data
- Data **visualization**: the graphical representation of information

Data science: extract forward-looking insights

Business Intelligence: descriptive analysis of historical data

Why do we visualize?

Where do we see visualization in real life?

What are the risks of using “messy” data?

Given an industry, what questions could data help answer?

Closing

- Check-out ticket
 - 3 things data science can do
 - 2 terms you learned today
 - 1 question you still have

What's next?

- The Data Science Lifecycle